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(54) **CHASSIS COMPONENT HAVING A
RECEPTACLE FOR AN ELASTOMER
BEARING**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2,925,974 A * 2/1960 Aebersold B60K 5/1216
267/134
3,072,448 A * 1/1963 Melton F16C 11/04
280/93.508
3,121,348 A * 2/1964 Reed B60G 7/001
74/588
4,202,564 A * 5/1980 Strader B60G 5/04
280/678
4,809,960 A * 3/1989 Kakimoto F16F 1/38
403/228

(Continued)

FOREIGN PATENT DOCUMENTS

CN 105599811 A * 5/2016 B60K 5/12
CN 106184369 A * 12/2016 B62D 21/02

(Continued)

OTHER PUBLICATIONS

Office Action for German Application No. 10 2022 107 298.3
mailed Nov. 18, 2022; 16pp.

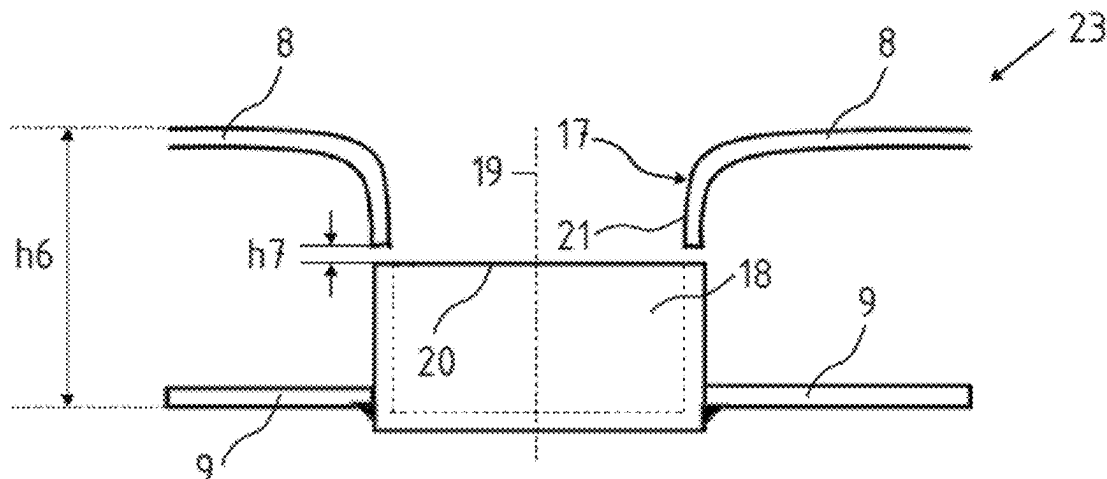
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(57) **ABSTRACT**

A chassis component having a receptacle for an elastomer bearing(s), the receptacle has a first cylinder section and a second cylinder section. The first cylinder section is formed by a rim hole and the second cylinder section is formed by a sleeve. The chassis component is formed from a first and a second sheet metal shell, and the sleeve is joined materially to the second sheet metal shell.

17 Claims, 2 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

5,695,213	A *	12/1997	Nakamura		B60G 7/001					9,908,572	B2 *	3/2018	Brooks		B62D 55/14
					280/124.134					10,179,609	B2 *	1/2019	Stenzenberger		B62D 21/11
6,109,654	A *	8/2000	Yamamoto		B62D 21/11					10,391,828	B2 *	8/2019	Jager		B62D 21/11
					280/124.109					10,953,719	B2 *	3/2021	Kiehn		B62D 21/11
6,241,267	B1 *	6/2001	Dzidosz		B60G 7/001					10,967,693	B2 *	4/2021	Meyer		B60G 7/001
					280/124.135					11,421,723	B2 *	8/2022	Maruyama		F16B 4/004
6,328,513	B1 *	12/2001	Niwa		F16F 1/362					11,466,714	B2 *	10/2022	Arndt		F16B 43/001
					411/339					11,999,213	B2 *	6/2024	Berger		B60G 7/001
6,547,267	B1 *	4/2003	Heep		B60G 7/001					2002/0140146	A1 *	10/2002	Nakagawa		F16F 1/3873
					280/124.128										267/141.2
6,796,765	B2 *	9/2004	Kosel		F02C 7/00					2007/0262503	A1 *	11/2007	Rogge		F16F 1/3863
					29/523										267/293
6,997,276	B2 *	2/2006	Yoshida		B62D 21/11					2015/0217808	A1 *	8/2015	Haselhorst		B62D 21/11
					180/312										280/124.109
7,100,264	B2 *	9/2006	Skinner		B23P 9/025					2017/0297398	A1	10/2017	Jager et al.		
					29/523					2018/0023604	A1 *	1/2018	Stastny		B62D 27/065
7,445,201	B2 *	11/2008	Endo		F16F 1/3863										403/376
					267/141					2019/0023322	A1 *	1/2019	Haselhorst		B62D 21/11
7,445,243	B2 *	11/2008	Park		B62D 21/11					2019/0061453	A1 *	2/2019	Branger		B60G 7/001
					280/124.109					2021/0309061	A1 *	10/2021	Sieve		B60G 7/001
7,682,117	B2 *	3/2010	Holt		F02M 55/025					2023/0028093	A1 *	1/2023	Protzmann		B60K 5/1216
					403/408.1										
7,726,674	B2 *	6/2010	VanDenberg		B60G 11/62										
					267/257										
7,874,059	B2 *	1/2011	Morrison		F16B 5/0266										
					29/889										
8,128,308	B2 *	3/2012	Skinner		B23P 9/025										
					29/523										
8,388,000	B2 *	3/2013	Hessing		B21D 53/88										
					280/124.134										
8,414,002	B2 *	4/2013	Yu		B60G 7/001										
					280/124.134										
8,777,196	B2 *	7/2014	Kumper		F16F 1/3814										
					267/293										
8,783,796	B2 *	7/2014	Thorson		B62D 55/092										
					305/138										
9,010,716	B2 *	4/2015	Kobori		F16F 1/3735										
					248/635										
9,422,003	B2 *	8/2016	Yun		B62D 21/11										
9,797,443	B2 *	10/2017	Spinella		F16C 27/063										

FOREIGN PATENT DOCUMENTS

CN	108622199	A *	10/2018		
CN	111284561	A *	6/2020		
CN	112078354	A *	12/2020		
CN	114056430	A *	2/2022		
CN	116534126	A *	8/2023		
DE	102005013133	A1 *	9/2006		
DE	102014222577	A1	5/2016		
DE	102015200352	A1	7/2016		
DE	102015208898	B3 *	8/2016		B62D 21/11
DE	102011052398	B4	11/2016		
DE	102015210915	A1	12/2016		
DE	102016107155	A1	10/2017		
DE	102016113021	A1 *	1/2018		B62D 21/11
DE	102019131716	B4	1/2022		
EP	4349692	A2 *	4/2024		B62D 21/11
WO	WO-2005123487	A1 *	12/2005		B62D 21/11
WO	WO-2005123488	A1 *	12/2005		B62D 21/11
WO	WO-2022064116	A1 *	3/2022		B62D 21/11

* cited by examiner

Fig. 1

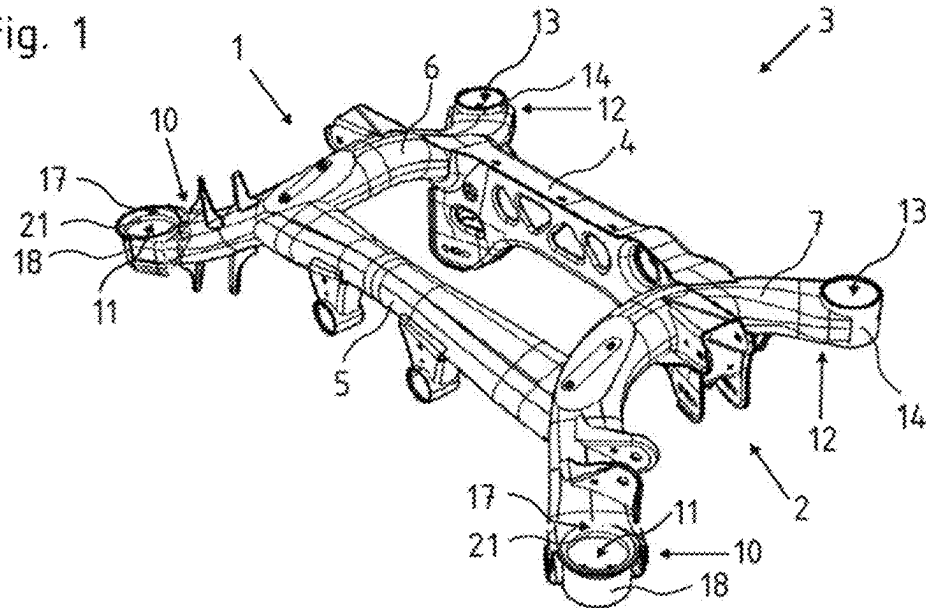


Fig. 2

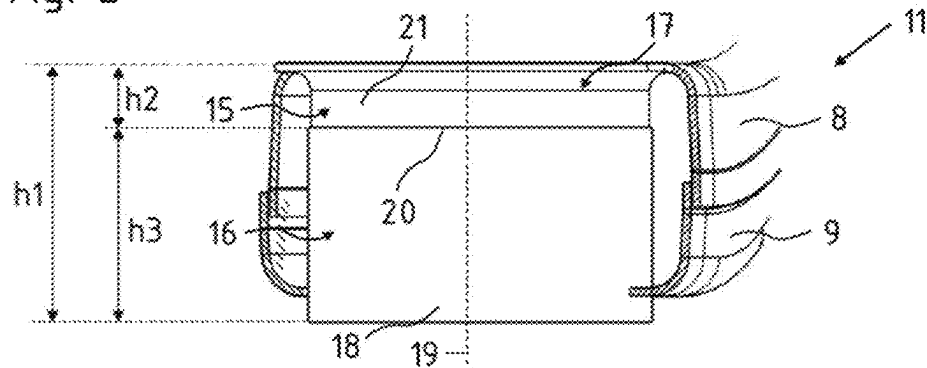


Fig. 3

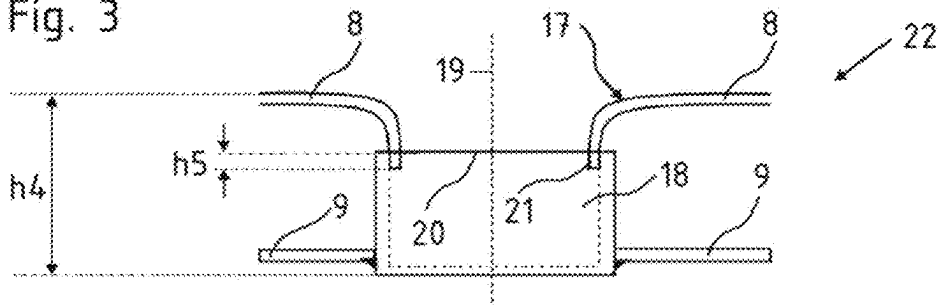


Fig. 4

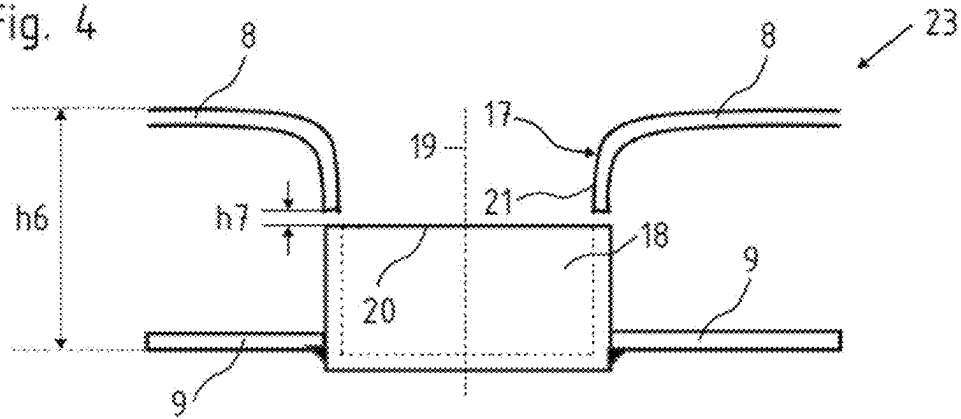


Fig. 5

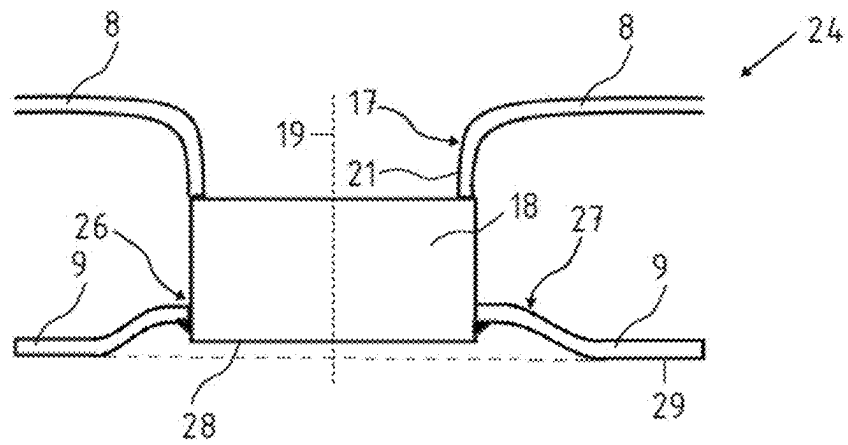
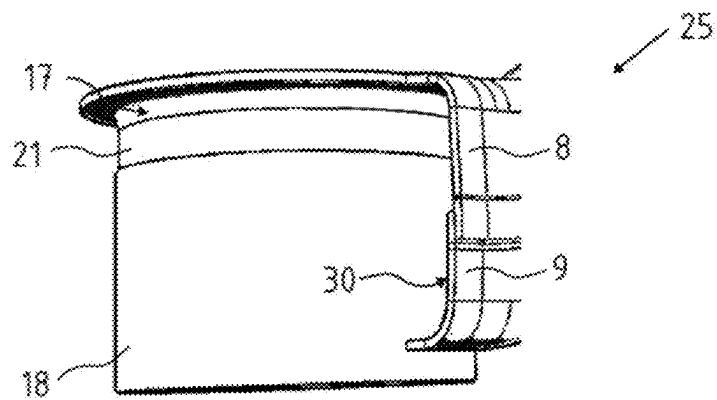


Fig. 6



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CHASSIS COMPONENT HAVING A RECEPTACLE FOR AN ELASTOMER BEARING

RELATED APPLICATIONS

The present application claims priority of German Application Number 10 2022 107 298.3 filed Mar. 28, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

FIELD

The disclosure relates to a chassis component.

The chassis is part of a motor vehicle. The chassis is responsible for passenger comfort and makes a contribution to driving dynamics and safety by ensuring that the forces generated in the contact zones between the road and the tire are transmitted in an optimum manner. The chassis is a complex combination of chassis components which serve to connect the body to the roadway.

BACKGROUND

In motor vehicles, chassis components are connected to the body, drive components or other chassis components via bearing(s). Elastomer bearing(s) are used in chassis assemblies for motor vehicles. They are mounted at the connection points to the vehicle body, to the transmission, to the drive unit or to chassis links in order, for example, to decouple the vibration or noises of the differential. Elastomer bearing(s) are able to absorb forces by deformation of the elastomer and are pressed by means of the outer sleeve into a correspondingly prepared receptacle of the chassis component.

Sleeves are used to receive the elastomer bearing(s), as described by DE 10 2019 131 716 B4. The sleeves are able to be made, for example, from a tube section or are able to be designed as a slotted or rolled sleeve. The sleeves are joined to the chassis component or are riveted or screwed to the chassis component in the form of separate parts, for example, as a forged part. In order to join the sleeves materially, a sufficient free space, e.g., for a welding unit, should be ensured and a corresponding projection for the formation of the welded seam should additionally be provided structurally. In the case of complex installation spaces, said configuration leads to design limitations.

Alternatively, the receptacle is able to be designed as a double-sided rim hole, as shown in DE 10 2016 107 155 A1 and DE 10 2011 052 398 B4. A rim hole in a workpiece is able to be formed, for example, by a punching, drilling, roller forming or spreading method. In the case of a rim hole, connection by means of joining is therefore eliminated. The execution of the rim hole is demanding in terms of production technology and limited to a maximum depth of the rim hole collars. This is a disadvantage compared with the sleeve solution, in which the depth of the receptacle is not limited. In the chassis sector, rim hole depths of between 18 mm and 20 mm are nowadays common. Elastomer bearing(s) are able to have a height of more than 60 mm. The rim hole thus limits the height of the elastomer bearing(s), or a distance is created between the rim holes which is able to adversely affect the secure retention of the elastomer bearing(s) and lead to manufacturing problems. Owing to the fact that the rim hole collars cannot be deformed to an arbitrary depth, the friction area with respect to the elastomer bearing(s) and thus the maximum effective press-out force is reduced.

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DE 10 2015 200 352 A1 describes a link for a wheel suspension of a vehicle having a receptacle for an elastomer body. This publication is able to be referred to as defining the type in question.

DE 10 2015 210 915 A1 also describes a link for a wheel suspension of a vehicle having a bearing(s) receptacle. The receptacle is formed by a bushing and two opposing rim holes.

DE 10 2014 222 577 A1 describes a control arm with a ball joint in press-fit connection.

SUMMARY

At least one embodiment of the disclosure provides a chassis component having an improved receptacle for an elastomer bearing(s), which ensures a large friction area between the elastomer bearing(s) and the receptacle and doing so with as few joining operations as possible, thus enabling the receptacle to be implemented in installation spaces with limited free space.

According to at least one embodiment of the disclosure, a chassis component includes a receptacle for an elastomer bearing(s).

A chassis component is, for example, a subframe, longitudinal member, crossmember or A arm. The receptacle of the chassis component has a first and a second cylinder section, wherein the first cylinder section is formed by a rim hole and the second cylinder section is formed by a sleeve.

The chassis component according to at least one embodiment of the disclosure is formed from a first and a second sheet metal shell. The two sheet metal shells are assembled to form the chassis component and joined to or along the contacting limbs. The sheet metal shells form a hollow profile, at least in some location or locations. The rim hole is a materially unitary part of a first sheet metal shell. According to at least one embodiment of the disclosure, the sleeve is joined materially to a second sheet metal shell. The sleeve is able to be designed, for example, as a slotted, perforated, multi-layered rolled or longitudinally joined sheet/tube.

The rim hole and the sleeve are arranged on a common central longitudinal axis. The end faces of the inner end of the sleeve and of the collar of the rim hole are situated opposite one another so as to rest against one another. The clearance is between 0.1 and 10 mm, between 0.2 and 5 mm and between 0.3 and 3 mm.

The structural length of the receptacle is able to be adapted to the elastomer bearing(s), such as by adapting the sleeve length or the depth of the rim hole. The receptacle is able to be adapted to the height of the elastomer bearing(s) by means of the distance between the end faces of the sleeve and the collar.

The combination of a sleeve and a rim hole ensures a large friction area of the receptacle with the elastomer bearing(s). At the same time, locations which are critical in terms of installation space since are able to be responded to, in comparison with the pure sleeve solution, one of the two joining operations is able to be omitted. The sleeve becomes shorter and thus less expensive than the pure sleeve solution. The rim hole is able to be executed at a depth that is non-critical in terms of production technology and cost-effective. The combination of a sleeve and a rim hole is able to be used for many receptacles of chassis components, such as for subframes and A arms and control arms.

In at least one embodiment of the receptacle, the sleeve is joined, for example welded, to the second sheet metal shell over an entire circumference of the sleeve in the region of an

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opening of the second sheet metal shell. The opening is circular or oval. Alternatively, the sleeve also is able be joined materially to the second sheet metal shell in the region of a part-circular opening of the second sheet metal shell, for example a semicircle, such as by means of a weld seam and at the end of a control arm leg. The opening of the second sheet metal shell is able to be arranged in a recess, thus ensuring that the outer end of the joined sleeve does not project beyond the surface of the second sheet metal shell, or terminates flush with the second sheet metal shell. In this context, the surface of the second sheet metal shell is understood to be the section around the elastomer bearing(s). This notwithstanding, the second sheet metal shell also is able to be designed to be recessed in a section at least 200 mm away from the elastomer bearing(s), for example, at an opposite end of the chassis component.

In at least one embodiment of the receptacle, the inner end of the sleeve and the collar of the rim hole are able to overlap in a certain region and, for example, are able to be connected to one another in a non-positive and/or positive manner. The receptacle is able to be adapted to the height of the elastomer bearing(s) by means of the height of the overlap between the sleeve and the collar.

Sheets of a steel alloy are used as the material for the chassis component. The sleeve is made of a material different from the sheet metal shells, but also of a steel material. The sheet metal shells also is able to have tailor-made properties, for example, in respect of mechanical characteristics or wall thicknesses. The chassis component is painted. The chassis component is able to have a corrosion protection coating.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure is described below with reference to drawings. In the drawings:

FIG. 1 shows a chassis component as part of a subframe in a perspective view according to at least one embodiment of the disclosure;

FIG. 2 shows a perspective front view of a receptacle according to at least one embodiment of the disclosure;

FIG. 3 shows a front view of the receptacle according to at least one embodiment of the disclosure;

FIG. 4 shows a front view of the receptacle according to at least one embodiment of the disclosure;

FIG. 5 shows a front view of the receptacle according to at least one embodiment of the disclosure; and

FIG. 6 shows a perspective view of the receptacle according to at least one embodiment of the disclosure.

DETAILED DESCRIPTION

FIG. 1 shows chassis components in the form of two longitudinal members 1, 2, which are part of a subframe 3. The longitudinal members 1, 2 are connected to one another by two crossmembers 4, 5.

The chassis components are formed from a main body 6, 7. The main body 6, 7 has two sheet metal shells 8, 9. The sheet metal shells 8, 9 are formed by pressing and are arranged one above the other in the image plane with the exception of overlapping joining regions.

A receptacle 11 according to at least one embodiment of the disclosure is arranged in each case at a rear end 10 of the main body 6, 7 in the direction of travel of the motor vehicle. A bearing(s) receptacle 13 is provided at a front end 12. In FIG. 1, the bearing(s) receptacles 13 are embodied as long sleeves 14. The long sleeves 14 are joined to the sheet metal

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shells 8, 9 of the main body 6, 7. According to at least one embodiment of the disclosure, the receptacles 11 have a first cylinder section 15 and a second cylinder section 16, wherein the first cylinder section 15 is formed by a rim hole 17 in the first sheet metal shell 8 and the second cylinder section 16 is formed by a sleeve 18. The sleeve 18 of the second cylinder section 16 is joined to the second sheet metal shell 9.

FIG. 2 shows the detail view of a receptacle 11 according to at least one embodiment of the disclosure for an elastomer bearing(s). The rim hole 17 and the sleeve 18 are arranged on a common central longitudinal axis 19. The inner end 20 of the sleeve 18 and the collar 21 of the rim hole 17 rest on the end face. The height h1 of the receptacle 11 is able to be adapted to the elastomer bearing(s) by means of the height h2 of the collar 21 of the rim hole 17 and the height h3 of the sleeve 18.

FIG. 3, FIG. 4, FIG. 5, and FIG. 6 show some embodiments of the receptacle 22, 23, 24, 25 for the elastomer bearing(s).

FIG. 3 shows at least one embodiment of the disclosure in the form of the receptacle 22, in which the inner end 20 of the sleeve 18 and the collar 21 of the rim hole 17 overlap in a certain region and are connected to one another in a non-positive manner. The height h4 of the receptacle 22 is able to be adapted to the elastomer bearing(s) by means of the height h5 of the overlap.

FIG. 4 shows at least one embodiment of the disclosure in the form of the receptacle 23, in which the end faces of the inner end 20 of the sleeve 18 and of the collar 21 of the rim hole 17 are situated opposite one another. The height h6 of the receptacle 23 is able to be adapted to the elastomer bearing(s) by way of the clearance h7 between the inner end 20 of the sleeve 18 and the collar 21 of the rim hole 17.

As shown in FIG. 5, at least one embodiment of the disclosure the opening 26 of the second sheet metal shell 9 is able to be arranged in a recess 27, thus ensuring that the outer end 28 of the joined sleeve 18 does not project beyond the surface 29 of the second sheet metal shell 9.

FIG. 6 shows at least one embodiment of the disclosure, in which the sleeve 18 is joined materially to the second sheet metal shell 9 by means of a semicircular weld seam in the region of a semicircular opening 30 of the second sheet metal shell 9.

The foregoing description of some embodiments of the disclosure has been presented for purposes of illustration and description. The description is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings. The specifically described embodiments explain the principles and practical applications to enable one ordinarily skilled in the art to utilize various embodiments and with various modifications as are suited to the particular use contemplated. Various changes, substitutions and alterations can be made hereto without departing from the spirit and scope of the disclosure.

The invention claimed is:

1. A chassis component, comprising:

a receptacle for an elastomer bearing;

a first sheet metal shell connected to the receptacle; and

a second sheet metal shell connected to the receptacle, the receptacle comprising a first cylinder section and a second cylinder section, the first cylinder section comprising a rim hole and the second cylinder section comprising a sleeve, wherein the sleeve is welded to the second sheet metal shell, wherein

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the rim hole projects inwardly from the first sheet metal shell in a direction towards the sleeve and the second sheet metal shell,

an end face of the sleeve and an end face of a collar of the rim hole are situated opposite one another,

the rim hole and the sleeve are on a common central longitudinal axis, and

along the common central longitudinal axis, an entirety of the sleeve is spaced from an entirety of the collar of the rim hole.

2. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell over a circumference in entirety of the sleeve at a region of an opening of the second sheet metal shell.

3. The chassis component according to claim 1, wherein the rim hole is a materially unitary part of the first sheet metal shell.

4. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell in a region of an opening of the second sheet metal shell.

5. The chassis component according to claim 4, wherein the opening of the second sheet metal shell is in a recess of the second sheet metal shell.

6. The chassis component according to claim 4, wherein an outer end of the sleeve does not project beyond a surface of the second sheet metal shell.

7. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell in a region of a part-circular opening of the second sheet metal shell.

8. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole overlap one another when viewed from an overhead perspective of the chassis component.

9. The chassis component according to claim 1, wherein the chassis component is a link.

10. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole are connected to one another in a non-positive manner.

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11. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole are connected to one another in a positive manner.

12. The chassis component according to claim 1, wherein the chassis component is a subframe of a motor vehicle or a part thereof.

13. The chassis component according to claim 1, wherein the chassis component is an engine frame of a motor vehicle or a part thereof.

14. The chassis component according to claim 1, wherein along the common central longitudinal axis, the end face of the sleeve faces toward, and is spaced from, the end face of the collar of the rim hole.

15. The chassis component according to claim 14, wherein the entirety of the sleeve is outside the rim hole.

16. A chassis component, comprising:

a receptacle for an elastomer bearing;

a first sheet metal shell connected to the receptacle; and a second sheet metal shell connected to the receptacle, the

receptacle comprising a first cylinder section and a second cylinder section, the first cylinder section comprising a rim hole and the second cylinder section comprising a sleeve,

wherein

the sleeve is welded to the second sheet metal shell,

an end face of the sleeve and an end face of a collar of the rim hole are situated opposite one another,

the rim hole and the sleeve are on a common central longitudinal axis, and

along the common central longitudinal axis, an entirety of the sleeve is spaced from an entirety of the collar of the rim hole.

17. The chassis component according to claim 16, wherein the sleeve is welded to the second sheet metal shell over a circumference in entirety of the sleeve at a region of an opening of the second sheet metal shell.

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